UNIVERSITI MALAYA

JABATAN KEJURUTERAAN MEKANIK

Department of Mechanical Engineering

LAPORAN PROJEK

Project Report

Sesi & Semester **Session & Semester**	Semester 2, 24/25
Kod & Nama Kursus **Course Code & Name**	KIG2020 Mechatronic II
Tajuk Projek **Project Title**	EcoSync - Smart, Energy-efficient Air Conditioning Controller
Nombor Kumpulan **Group Number**	G1-B
Demo **Demonstrator**	Dr. Zhou Ding
Tarikh Hantar **Submission Date**	18 June 2025

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Project Title: EcoSync - Efficiency in sync with comfort.

1.0 Team Members and Responsibilities

Team Member	Matrix Number	Roles and Responsibilities
Mohamad Firdaus Hadzwan Bin Mohamad Syarul	22001684	Leader & Editor • Assign tasks and manage datelines • Final edit, formatting, submission
Aiman Harisya Bin Zaharudin	22001697	Functionality Tester and Demo Editor • Build and test prototype • Capture image of working system • Explain test results • In charge video Demonstration
Joaqim Elfian bin Mansor	22001625	Researcher and Writing • Draft introduction and objectives • Research project related and summary
Muhammad Aiman Bin Mohd Zaki	22001650	Circuit Designer • Draw and simulate in TinkerCAD • Explore components functionality
Irfan Shah Putra Bin Saiful Faizd	23065450	Reference and Supporter • Compile all references • Cite sources

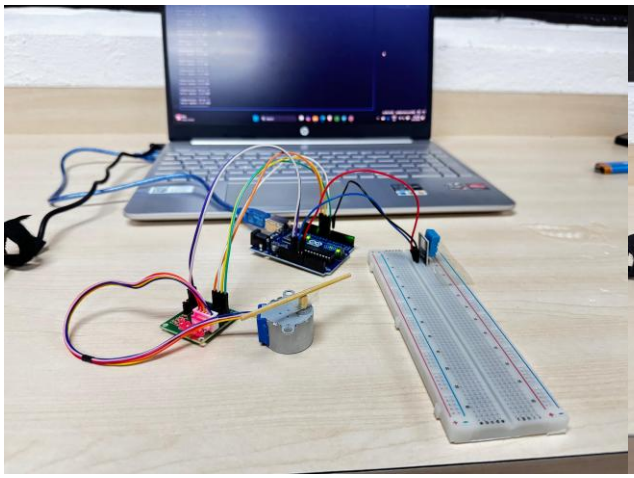
2.0 Introduction

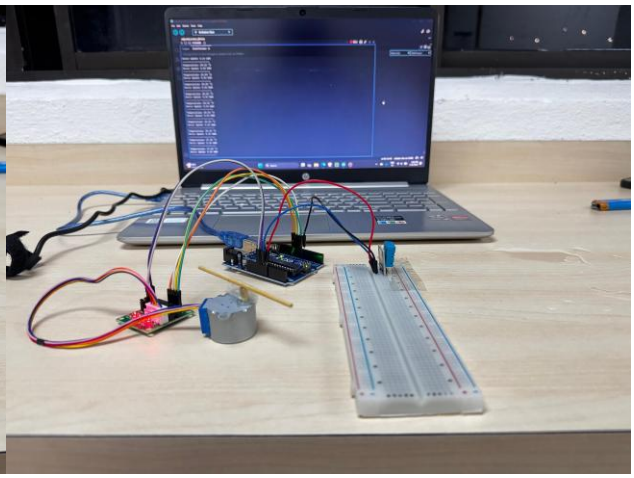
Air conditioning systems are one of the crucial things for indoor thermal comfort especially in tropical regions like Malaysia. However, typical air conditioning devices that run in a fixed ON/OFF mode always led to energy inefficiency and excessive electricity consumption. This not only will increase operational cost, but it also contributes to higher carbon emissions. Therefore, to support University Malaya Sustainability Policy 2021–2030, there is an increasing demand for smarter and more sustainable energy management systems within campus buildings.

This project proposes the development of smart temperature responsive air conditioning system where it is utilizing Arduino microcontroller, temperature sensor DHT11 and a DC motor (as a simulated compressor) . The device will automatically adjust the motor speed depending on real time ambient temperature data. The motor will speed up when the temperature rises above 32°C and will slowing down below 26°C. This mimics the behaviour of a variable speed compressor which is known to be more energy efficient than typical system.

Objective

- To develop an automatic temperature control system that maintains an optimal room climate by adjusting the air conditioning compressor speed based on real-time temperature readings.
- To improve energy efficiency and reduce carbon emissions by minimizing unnecessary ON/OFF cycles through continuous control of motor speed using Arduino and stepper motor simulation.





3.0 Understanding the Circuit Drawing

To understand the mechanism of a smart, temperature-responsive air conditioning system, three essential components were utilized: an Arduino microcontroller, a DHT11 temperature sensor, and a DC motor using simulation in TinkerCad. These elements were combined to replicate the functionality of a compressor in a conventional air conditioning unit. This simplified model serves as a functional prototype to demonstrate how temperature regulation can be achieved using basic electronics, simulating the smart control behaviour of modern air conditioners.

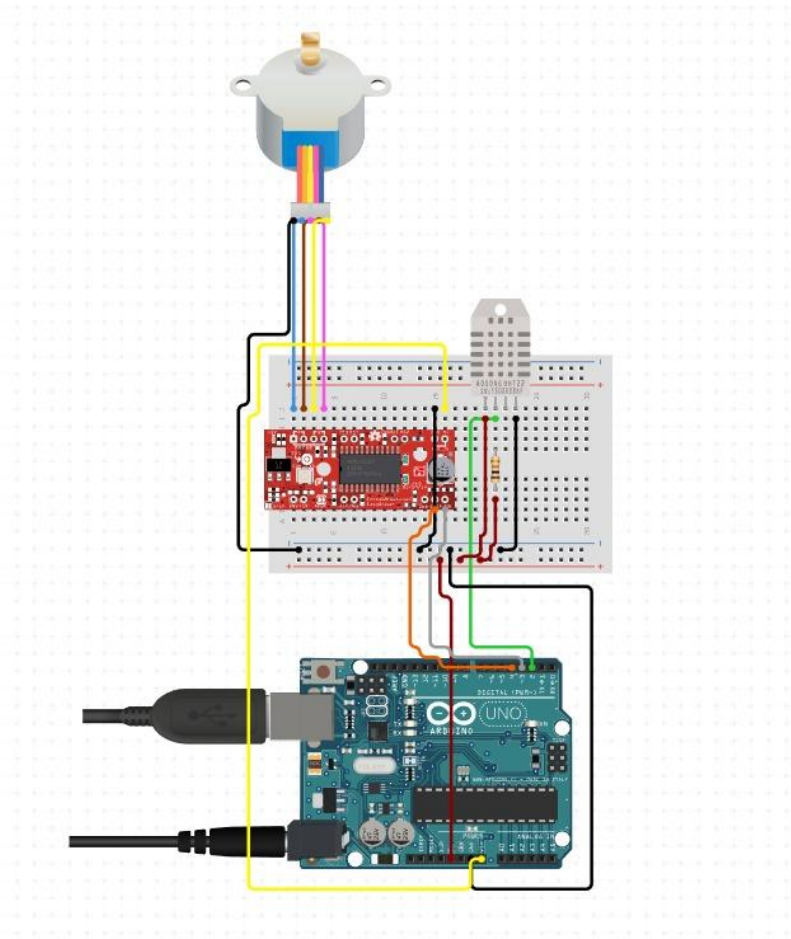


Diagram 1

Component:

1. Arduino Uno R3
2. DHT11 sensor
3. stepper DC motor
4. ULN2003 motor driver
5. breadboard
6. jumper wire

Procedures for Connection

1. DHT11 sensor was connected to the breadboard
   - VCC was connected to Vin Arduino
   - Data was connected to pin 7 Arduino
   - GND was connected to GND Arduino

2. Step motor
   - Five jumper wire was connected directly ULN2003 Stepper Motor Driver

3. ULN2003 Stepper Motor
   - GND was connected to GND Arduino
   - VCC was connected to 5V Arduino
   - IN1 was connected to pin 8 Arduino
   - IN2 was connected to pin 10 Arduino
   - IN3 was connected to pin 9 Arduino
   - IN4 was connected to pin 11 Arduino

The Arduino microcontroller serves as the control centre for this prototype. It processes input data from the DHT11 sensor, applies logical conditions, and controls the operation of the DC motor. The DHT11 is a digital temperature and humidity sensor that provides real-time temperature readings to the Arduino.

In this setup, when the ambient temperature detected by the DHT11, exceeds the defined threshold, the Arduino increase the speed of the motor. This simulates the compressor turning on to cool the environment. Once the temperature decrease, the Arduino slow down the motor, simulating the compressor slow down to conserve energy. This process repeats in a loop, continuously monitoring and regulating the temperature.

For example, the set point is 32°C with a small buffer zone to prevent the system from toggling on and off too frequently. If the temperature rises above 32°C, the motor increase it speeds. If it drops below 26°C, the motor is slow down. This approach ensures stable operation and prevents rapid cycling, which could damage a real compressor.

To implement this system, the DHT11 sensor is connected to a digital input pin on the Arduino, while the motor is connected to a digital output pin through a transistor or motor driver circuit. Both TinkerCAD simulation and real test produce desired output.

4.0 Arduino Coding

```
#include <Adafruit_Sensor.h>
#include <DHT.h>
#include <DHT_U.h>
#include <Stepper.h>

#define DHTPIN 7          // DHT11 sensor data pin
#define DHTTYPE DHT11     // DHT sensor type

// Stepper motor setup
const int stepsPerRevolution = 2048;  // For 28BYJ-48 stepper

// IN1=8, IN2=10, IN3=9, IN4=11
Stepper stepper(stepsPerRevolution, 8, 10, 9, 11);

// Temperature thresholds
#define TEMPERATURE_THRESHOLD 26
#define TEMPERATURE_THRESHOLD1 32

DHT dht(DHTPIN, DHTTYPE);

float currentRPM = 0;

void setup() {
  Serial.begin(9600);
  dht.begin();

  stepper.setSpeed(5); // Initial speed
  currentRPM = 5;
}

void loop() {
  float temperature = dht.readTemperature();

  if (isnan(temperature)) {
    Serial.println("Failed to read temperature from DHT sensor!");
    return;
  }

  // Control stepper motor based on temperature
  if (temperature > TEMPERATURE_THRESHOLD1) {
    currentRPM = 11;
    stepper.setSpeed(currentRPM);
    stepper.step(stepsPerRevolution); // One full rotation
```

```
  }
  else if (temperature > TEMPERATURE_THRESHOLD) {
    currentRPM = 8;
    stepper.setSpeed(currentRPM);
    stepper.step(stepsPerRevolution / 2); // Half rotation
  }
  else {
    currentRPM = 3;
    stepper.setSpeed(currentRPM);
    stepper.step(stepsPerRevolution / 4); // Quarter rotation
  }

  // Print temperature and motor RPM
  Serial.print("Temperature: ");
  Serial.print(temperature);
  Serial.println(" °C");

  Serial.print("Motor Speed: ");
  Serial.print(currentRPM);
  Serial.println(" RPM");
  Serial.println("--------------------");
}
```

5.0 Justification of Functionality

The testing of the automatic temperature control system device has done at different situations. At first, the device was tested at a room temperature before being cooled manually using an ice cube. The other test was done to test the functionality of the device to operate in a hot temperature (above the room temperature) by providing fire near the thermal sensor. Upon testing the prototype, there are few observations and justifications that can be made.

Results and Discussion:

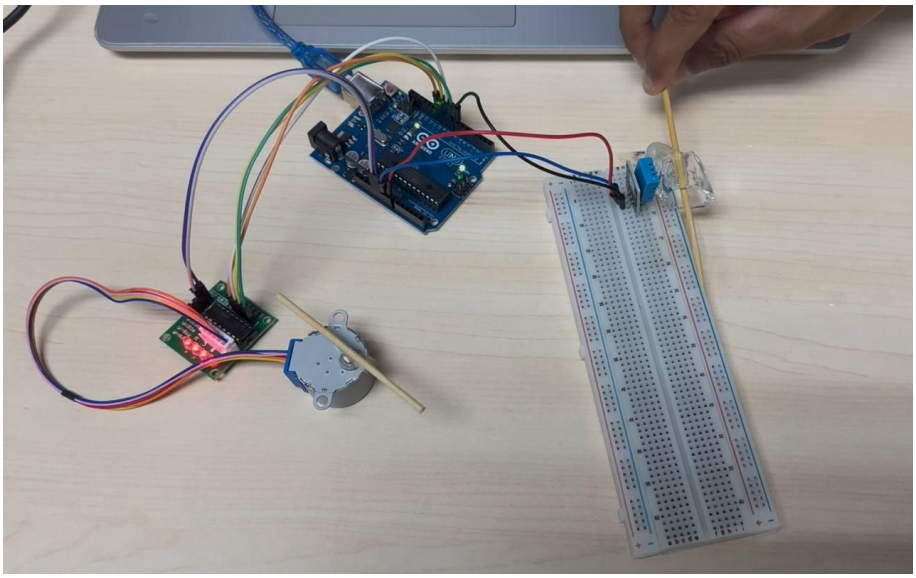


Figure 4.1: Test 1: Device Operating Under Room Temperature

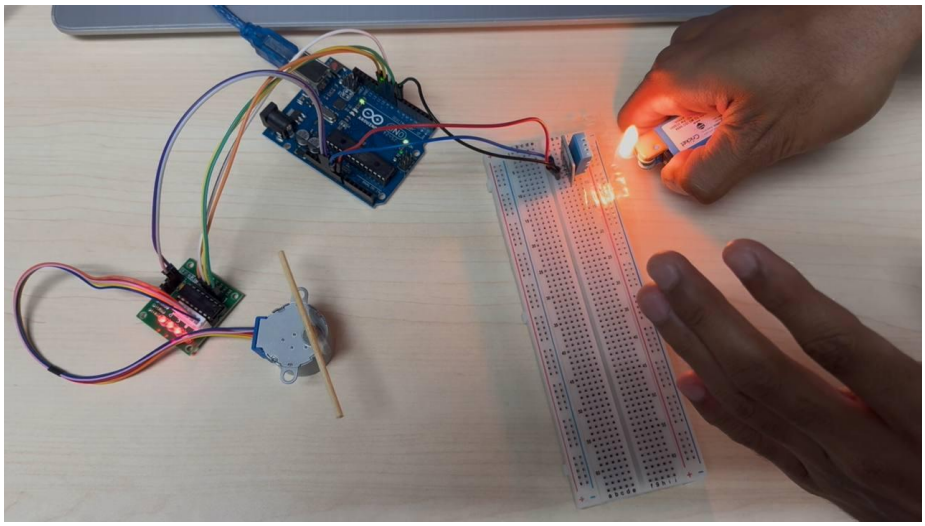


Figure 4.2: Test 2: Device Operating Above Room Temperature

```
Temperature: 26.20 °C
Motor Speed: 8.00 RPM
--------------------
Temperature: 25.60 °C
Motor Speed: 3.00 RPM
--------------------
```

Figure 4.3: Display of Test 1 Results

```
Temperature: 31.90 °C
Motor Speed: 8.00 RPM
--------------------
Temperature: 32.20 °C
Motor Speed: 11.00 RPM
--------------------
```

Figure 4.4: Display of Test 2 Results

At normal temperature, the stepper motor operates at a medium speed of 8 RPM in both tests. However, when the temperature recorded by the thermal sensor drops below 26°C, the motor slows down to 3 RPM. This reduction in speed occurs because the surrounding temperature is already cooler than the desired level, requiring less effort from the motor. Conversely, in the second test, the motor runs at a higher speed of 11 RPM due to the ambient temperature being higher than the target temperature which is 32°C. In this case, the motor needs to work harder to cool the environment to the optimal level. The motor's ability to adjust its speed based on ambient temperature demonstrates the system's efficiency in optimizing energy use and reducing carbon emissions, aligning well with the objectives of this project.

Looking ahead, future improvements could focus on integrating machine learning algorithms to further optimize speed control based on predictive temperature patterns. Additionally, enhancing sensor accuracy, expanding wireless connectivity, and incorporating renewable energy sources like solar panels could significantly boost the system’s sustainability and adaptability. These developments would not only improve overall performance but also support broader environmental goals by further minimizing energy waste and carbon footprint.

6.0 Summary

The EcoSync project aims to create a smart, energy efficient air conditioning control using Arduino, DHT11 temperature sensor, and a simulated DC motor compressor, and to have in place a dynamic control system to adjust the motor speed as the temperature changes, excess energy planned for reduction based on the absence of allowing “wasted” or carbon-producing energy. The finished product was tested in both cool and hot condition, and in both scenarios, with full control, the EcoSync system successfully adjusted the motor speed to maintain desired climate in the average indoor room temperature. There was measurable improved energy efficiency in the use of humidity and temperature sensor monitoring motor speeds with the EcoSync system, which aligned with University Malaya’s goals in sustainability and conservation. EcoSync could possibly evolve into an upgraded system capable of machine learning, harnessing solar energy or operating wirelessly.

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